INCH-POUND

MIL-C-24643/25D
22 November 1994
SUPERSEDING
MIL-C-24643/25C
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight with circuit integrity)

First - Copper conductor, nickel coated, AWG 18, class B stranding.
* Second - Extruded Silicone rubber insulation. Minimum average wall thickness 0.018 inch.
Third - Glass braid, (one colored black and one colored white for each pair of conductors).
* Fourth - Transparent braid covering. Two conductors (one black and one white) cabled together to form a pair with lay not greater than 2.25 inches.
Fifth - Braided shield of AWG No. 34 or 36 uncoated copper
* over each pair. Minimum coverage of 85 percent.
Sixth - Shield insulation over each braided shield, consisting of polyethylene terephthalate film, type G in accordance with MIL-I-631, 0.001 inch thick, applied helically with 50 percent overlap.
Seventh- A glass braid over each shield insulation. (Standard identification code applied by method 4.)
Eighth - The specified number of pairs (see table I) cabled together with a lay not greater than 14 inches. Fillers saturated with non-charring silicone base compound shall be used as necessary, to form a firm, round cable.

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Construction (non-watertight with circuit integrity) (continued)

Ninth - A binder tape applied helically with overlap.
Tenth - Silicone rubber jacket colored orange or red. (see table I for thickness).
Eleventh - Braided metal armor

* TABLE I. Details

Military part no. M24643/25	Type and Size	Number of pairs	Cable jacket thick-ness min avg (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01AN	LSPI-3	3	0.050	0.685	9.0
-02AN	LSPI-7	7	.063	.900	11.0
-03AN	LSPI-12	12	.063	1.155	14.0

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.5
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	1200
Shield to shield	500
Insulation resistance - megohms/1000 feet, minimum...	
* Conductor to shield	500
* Shield to shield	100
* Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
The nickel coating on the conductors of the completed cable shall not be nicked, scraped or abraded so as to expose the copper conductor	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas flame - 1 hour at 120/70 volt root mean square	No failure
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	800
Elongation - percent, minimum	250

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EXAMINATION AND TESTS (continued):	Requirements
Durometer (type A) hardness, minimum (ASTM D 2240)	60
Group C:	
Physicals - cable aging (260 ±10°C)	
Jacket	
Tensile strength - lb/in2, minimum	600
Elongation - percent, minimum	150
Group D:	
* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Gas flame (3 hours) at 120/70 volts RMS	No failure
Aging and compatibility (cable) (125 ±5°C.) For this test, omit all references to water-tightness and fillers.	No failure
Armor - conformance to material, construction and coverage	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
* Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
* Smoke index, maximum	
Jacket	35
Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5

UNIT ORDERING LENGTH:

All types and sizes 100 feet minimum.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - SH

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(Project No. 6145-2057-25)

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